Ahmed Hannan

Software Engineer & Computer Science Graduate

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Professional Summary

Computer Science senior at FAST-NUCES (graduating June 2026) with production experience in full-stack development and applied work in deep learning and high-performance computing. Built production cross-platform mobile apps in Flutter and web applications in React at EmpowerBits, conducted generative AI research at the National Center of AI (NCAI), and engineered systems spanning custom CUDA kernels, multimodal RAG pipelines, and parallel graph analytics on distributed clusters.

Education

National University of Computer & Emerging Sciences (FAST-NUCES), BS in Computer Science — Graduation: June 2026

- Relevant Coursework: Distributed Systems, High-Performance Computing, Deep Learning, Generative AI, Machine Learning, Computer Vision, Advanced Operating Systems, Database Systems, Software Engineering, Computer Networks, Design & Analysis of Algorithms, Linear Algebra.
- Academic Appointment: Selected as Undergraduate Teaching Assistant for Programming Fundamentals (C/C++), instructing lab sessions and mentoring 50+ students in manual memory management and data structures.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, Dart, SQL, Bash, CUDA C/C++

AI & Machine Learning: PyTorch, TensorFlow, HuggingFace Transformers, Sentence-Transformers, Scikit-Learn, OpenCV

Generative AI & LLMs: Multimodal RAG, FAISS Vector Search, Prompt Engineering, Fine-Tuning (LoRA, PEFT), Gemma 3, BERT, spaCy

Frontend & Mobile: React.js, Next.js, Flutter (iOS & Android), Tailwind CSS, Redux Toolkit, State Management, Vite

Backend & APIs: Node.js, Express.js, Django REST Framework, FastAPI, Flask, RESTful APIs, WebSockets, Redis

Databases & Cloud: PostgreSQL, MongoDB, SQLite, Google BigQuery, Firebase, Cloud SQL

Systems & HPC: CUDA Kernels, Tensor Cores, OpenMP Multithreading, MPI Distributed Computing, Linux, Socket Programming

DevOps & Tooling: Git/GitHub, Docker, CI/CD Pipelines, Jira, Agile/Scrum, Selenium WebDriver, Apache Airflow

Professional Experience

EmpowerBits, Full-Stack Developer — June 2026 – Present

- Developed cross-platform mobile applications in Flutter (iOS and Android) and responsive web dashboards in React for production client platforms.
- Engineered backend REST APIs using Node.js, Express, and PostgreSQL, implementing connection pooling and database indexing for high concurrency.
- Implemented server-side Redis caching and optimized database query plans, reducing median API response times by 30%.
- Architected clean reactive UI state management with Provider and Riverpod, reducing client-side widget rebuilds and eliminating memory leaks.
- Received the EmpowerBits Certificate of Excellence for technical ownership and on-schedule feature delivery across client release sprints.

RedCoast, Full-Stack Software Engineer — June 2025 – June 2026

- Built client-facing web and mobile features using React, Django REST Framework, Flutter, and PostgreSQL across multiple production releases.
- Collaborated directly with client teams during bi-weekly sprints to gather technical requirements, scope architectural trade-offs, and draft API schemas.
- Automated cross-browser regression testing suites using Selenium WebDriver and Jira, reducing manual release verification cycles.
- Constructed SQL aggregation pipelines in Google BigQuery to process user engagement logs, identifying usage trends and system performance bottlenecks.

National Center of AI (NCAI, NUST), AI Research Intern — August 2024 – September 2024

- Researched generative transformer architectures and vision-language foundation models, studying text-to-image synthesis pipelines and cross-attention mechanics.
- Benchmarked model inference speed and GPU VRAM footprint across FP16 and FP32 precisions on NVIDIA hardware.
- Authored technical documentation analyzing computational requirements, parameter scaling trade-offs, and deployment viability for on-premise hardware.

FAST – NUCES, Teaching Assistant (Programming Fundamentals) — June 2024 – September 2024

- Conducted weekly lab sessions, code reviews, and office hours for 50+ undergraduate students in C/C++ Programming Fundamentals.
- Taught core systems programming concepts including manual memory management (malloc/free), pointer arithmetic, recursion, and struct memory layout.
- Assisted students in identifying memory leaks, buffer overflows, and segmentation faults using debugging tools including GDB and Valgrind.

Featured Engineering & Research Projects

Pose2Play — Real-Time VR & Computer Vision Rehabilitation Platform (FYP) GitHub: Pose2Play

- Developed an interactive physical therapy platform combining real-time computer vision pose estimation with an immersive Unity VR environment.
- Fine-tuned a YOLOv8-pose model to detect 17 skeletal joint keypoints, calculating biomechanical metrics (knee flexion, shoulder abduction) in real-time.
- Engineered a low-latency socket IPC bridge streaming 3D joint coordinates from the Python inference server to Unity C# at 30+ FPS (<40ms latency).
- Implemented visual guidance feedback within VR to show target posture alignments and notify patients of improper form during exercise routines.

Multimodal Document RAG & Cross-Encoder Semantic Search GitHub: multimodal-rag-semantic-search

- Engineered an end-to-end multimodal retrieval-augmented generation pipeline to extract, parse, index, and query complex financial PDF filings.
- Extracted text, tabular structures, and embedded visual charts using PyMuPDF and Tesseract OCR, generating dense vectors via CLIP-ViT-B/32 (512-dim) and all-MiniLM-L6-v2 (384-dim) stored in FAISS.
- Deployed quantized Google Gemma-3-1B-IT locally with FP16 precision and layer offloading, executing grounded inference within a 3 GB GPU VRAM constraint.
- Fine-tuned a BERT-Tiny cross-encoder model on 1.1 million query-document pairs from the Amazon ESCI dataset to rerank search candidates by semantic relevance.

CRAG: Corrective & Multi-Pipeline RAG Benchmark Platform GitHub: crag-advanced-rag-pipelines

- Implemented and benchmarked 7 retrieval pipelines—Standard RAG, RAG Fusion, HyDE, Corrective RAG (CRAG) with confidence gating, Graph RAG, and RRR—on the Meta CRAG noisy web corpus.
- Developed a modular evaluation harness using spaCy named entity recognition, measuring exact match accuracy, retrieval hallucination rate, and execution latency.
- Built a Flask REST API backend integrated with FAISS vector indexing and a responsive React + Vite frontend dashboard for side-by-side pipeline inspection.

Parallel Implementation of ParButterfly on Large Bipartite Graphs HPC Graph Analytics

- Designed and implemented a parallel counting algorithm for 4-vertex butterfly motifs on billion-edge bipartite graphs using OpenMP (shared memory) and MPI (distributed memory).
- Optimized vertex degree sorting, compressed sparse row (CSR) graph storage, and L1/L2 cache locality, minimizing thread synchronization stalls.
- Evaluated scalability on symmetric multiprocessing (SMP) and cluster environments, achieving near-linear speedup across 16 CPU cores on large web graph benchmarks.

GPU Neural Network Kernel Acceleration (CUDA C++) GitHub: Neural-Network-Acceleration

- Wrote custom CUDA C++ kernels to accelerate general matrix multiplication (GEMM) and 2D convolution forward and backward passes on NVIDIA GPUs.
- Implemented 2D shared-memory tiling, register caching, and coalesced global memory access patterns to maximize memory bus throughput.
- Profiled warp occupancy, register pressure, and shared memory bank conflicts using NVIDIA Nsight Compute, achieving performance competitive with cuBLAS.

GoMatch — Real-Time Geolocation Ride Sharing Platform GitHub: GoMatch

- Built a full-stack mobile ride-sharing application with Flutter, Node.js, Express, and MongoDB implementing passenger hailing and driver matching.
- Integrated live background GPS tracking and WebSocket channels for bidirectional low-latency driver location updates and route rendering.
- Engineered transactional ride state management (request, match, pickup, transit, completion) with JWT authentication and fare estimation logic.

Enterprise Field Operations Mobile Suite GitHub: enterprise-field-operations

- Built an offline-first enterprise mobile application in Flutter for field distribution logistics, order lifecycles, and inventory catalog navigation.
- Engineered background synchronization using local SQLite transactions, automatically reconciling order queues and telemetry when network connectivity is restored.

Honors, Awards & Professional Activities

- Certificate of Excellence / Performance Certificate: Awarded by EmpowerBits for high-impact production engineering and timely milestone delivery.
- Undergraduate Teaching Assistantship: Appointed by FAST-NUCES Faculty of Computer Science for Programming Fundamentals based on technical merit.
- Open-Source Contributions: Actively maintain 18+ public GitHub repositories covering deep learning pipelines, GPU acceleration, systems programming, and mobile applications.